One Modular Block Refutes the Half-Count Law for Involution Centralizers

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AI-assisted disproof of conjecture 00000005838

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Abstract

Both source languages assert that an involution centralizer has half of its blocks real-valued. Take the genuine involution in $G = C_2$ and the prime $p = 2$. Over $K = \overline{\mathbb{F}_2}$ the actual modular group algebra $K[C_G(s)]$ has only the idempotents zero and one, and therefore precisely one primitive central idempotent, or one modular block. Any selected subfamily of this singleton has an integer cardinality, which cannot equal $1/2$. We formalize this stronger predicate-parametric contradiction: no definition of a predicate selecting blocks can make the asserted half count true for this example. No ordinary-character classification or invented realness test is required. All group, centralizer, field, algebra, idempotence, primitivity, cardinality and half-count bridges are explicit.

1 Exact Necessary Clause and Standard Block Convention

The English source includes the clause "the number of real-valued blocks is half the total"; the Chinese source asserts the same half-count equality. It is a separate necessary conjunct, not a conditional on an unstated hypothesis about the surrounding dichotomy. Neither language requires odd p, a nonabelian group or a simple group. We refute this half-count clause and do not formalize or repair the source's undefined kernel/dichotomy language, or separately refute its conjugate-pair clause.

The fixed complete bilingual source is retained separately:

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Commit:  b355e42906e6eaf52bc7e9d005e0ef02d8aa7400
Blob:  bcf128f6c8b7e7c438e4e096e9e99e38c3135082.
```

For a finite group H and an algebraically closed field K of characteristic p, we use the standard modular algebra definition of its p-blocks: nonzero primitive central idempotents of $K[H]$. Thus an idempotent e is central, satisfies $e^2 = e$, and is primitive when it cannot be written as a sum of two nonzero orthogonal central idempotents. This is the standard block decomposition convention, with its connection to the ordinary-character partition described in [1, Chapter 12, Sections 12.1–12.2]. We do not instead count central idempotents of the characteristic-zero semisimple algebra $\mathbb{C}[H]$.

The Lean block type is *defined* by these actual ring equations and the indecomposability condition. The general correspondence to an ordinary character-partition API is cited and reviewed, not advertised as a Lean character-theory theorem. No such theorem or realness classification is needed for a contradiction applying to every predicate on the actual modular block type. This standard algebraic definition, and not an arbitrary Boolean labeling, fixes what is counted below.

2 An Actual Involution Centralizer and Splitting Field

Let $G = C_2 = \{1, s\}$, with $s \neq 1$ and $s^2 = 1$. It is a finite group of order two. Since it is commutative,

$$C_G(s) = G.$$

These are genuine group and subgroup statements, not an empty involution case. In Lean, G is `Multiplicative (ZMod 2)` and the centralizer is the actual Mathlib subgroup of elements commuting with s. The explicit group isomorphism $C_G(s) \simeq G$ induces an actual algebra isomorphism $K[C_G(s)] \simeq K[G]$ by reindexing the group basis. Idempotence and zero/one are transported across that isomorphism.

Take $p = 2$ and the concrete field $K = \mathrm{AlgebraicClosure}(\mathbb{F}_2)$. Its field, characteristic-two and algebraic-closedness instances are constructed in Mathlib and explicitly audited here. The computation in the next section works over *every* characteristic-two field, so it is valid over this actual splitting field, not just an enumeration over the two-element field.

3 All Idempotents of the Modular Group Algebra

Write $[g]$ for the actual group-algebra basis vector at g. Every element x of $K[C_2]$ has a unique coefficient decomposition

$$x = a[1] + b[s], \qquad a, b \in K.$$

This is proved for every group-algebra element using its actual finite coefficient function and the two group elements. The multiplication rule is $[g][h] = [gh]$, hence, in characteristic two,

$$x^2 = (a^2 + b^2)[1] + (ab + ba)[s] = (a^2 + b^2)[1].$$

If $x^2 = x$, comparison of the $[s]$ coefficients first gives $b = 0$. Comparison at $[1]$ then gives $a^2 = a$. Since K is a field, $a(a-1) = 0$, so $a = 0$ or $a = 1$. Consequently

$$\forall x \in K[C_2], \quad x^2 = x \implies x = 0 \text{ or } x = 1.$$

This is a universal proof over the actual infinite splitting field, not a conclusion from a finite grid or a stored count. Transport along the centralizer algebra isomorphism gives the same result in $K[C_G(s)]$.

Lemma 1. *If a nontrivial ring has only the idempotents zero and one, then it has exactly one nonzero primitive central idempotent.*

Proof. One is nonzero, central and idempotent. Suppose it decomposes into orthogonal central idempotents f, g. If either is zero, the decomposition is trivial. Otherwise both must equal one by the hypothesis, contradicting $fg = 0$. Thus one is primitive. Every nonzero primitive central idempotent is a nonzero idempotent and hence equals one. This proves an actual singleton block type, including existence, uniqueness and finiteness. □

Applying the lemma to $A = K[C_G(s)]$ yields

$$|\operatorname{Bl}(A)| = 1.$$

The ring is nontrivial because the coefficient field is nontrivial and the group basis is nonempty. Lean proves finiteness of the full block type and each predicate subtype; no infinite-type default cardinality is used to obtain the count.

4 No Selection of Blocks Can Satisfy the Half Count

Theorem 1. *For the above genuine involution centralizer, no predicate P on its modular blocks satisfies*

$$|\{b \in \operatorname{Bl}(A) : P(b)\}| = \frac{|\operatorname{Bl}(A)|}{2}.$$

This holds with the equality interpreted in either $\mathbb{Q}$ or $\mathbb{R}$.

Proof. The selected subtype has a natural cardinality r. Since the full block count is one, the equality would imply $2r = 1$, impossible for any $r \in \mathbb{N}$. No value of P has been assumed, chosen or hard-coded. □

In particular this applies to the actual real-block predicate, whether real blocks are described as conjugation-stable blocks or via the corresponding character family. It is unnecessary to decide that predicate or classify the ordinary characters: the theorem excludes *every* possible block-selection predicate at once. The interpretation of half is the ordinary rational/real equality, not truncated natural division by two.

The formal final theorem allows an arbitrary selection family assigning a predicate to the actual blocks of the centralizer of each $t \in C_2$. It negates the half-count clause quantified over all nonidentity involutions, by evaluating that clause at the displayed s. Both its nonidentity premise and square-one premise are proved. Thus no arbitrary choice of realness can make this necessary clause valid for this finite group, and the source's universal conjunction is false. This argument does not claim that other source clauses are separately false.

5 Formal Scope and Reproducibility

The four mathematical modules are BlockTheory, Involution, C2Algebra and Counterexample. Their 34 public theorems include all coefficient, idempotence, field, involution, centralizer/algebra transport, finite cardinality and natural/rational/real half-count bridges. The final declaration is `Counterexample5838.half_clause_false_for_every_selection`. Its only explicit parameter is the arbitrary selection family; the conclusion is its negation of the actual necessary clause. No desired block count, realness value or negative conclusion is assumed.

Lean is pinned to 4.33.1 and Mathlib to

```
0df444a360eaa60ab8c11dca51a86af692955474.
```

Every public theorem's compiled type and axiom footprint is audited. Only the standard foundational axioms `propext`, `Classical.choice` and `Quot.sound` occur. Finite exact group-algebra checks are regression evidence, not the universal proof over K. Installed Lean and pinned dependency build caches remain explicit trust inputs; no full-dependency clean build, empty-machine bootstrap, cross-platform run or hosted-CI result is implied. Separate AI review is not human expert endorsement, upstream acceptance, priority, ranking or an award.

References

[1] P. Webb, *A Course in Finite Group Representation Theory*, author-hosted prepublication text, Chapter 12, printed pp. 226–232, especially Section 12.1, Proposition 12.2.1 and Example 12.2.2.